
机器人学

第三章：位姿描述和齐次变换

厦门大学信息学院

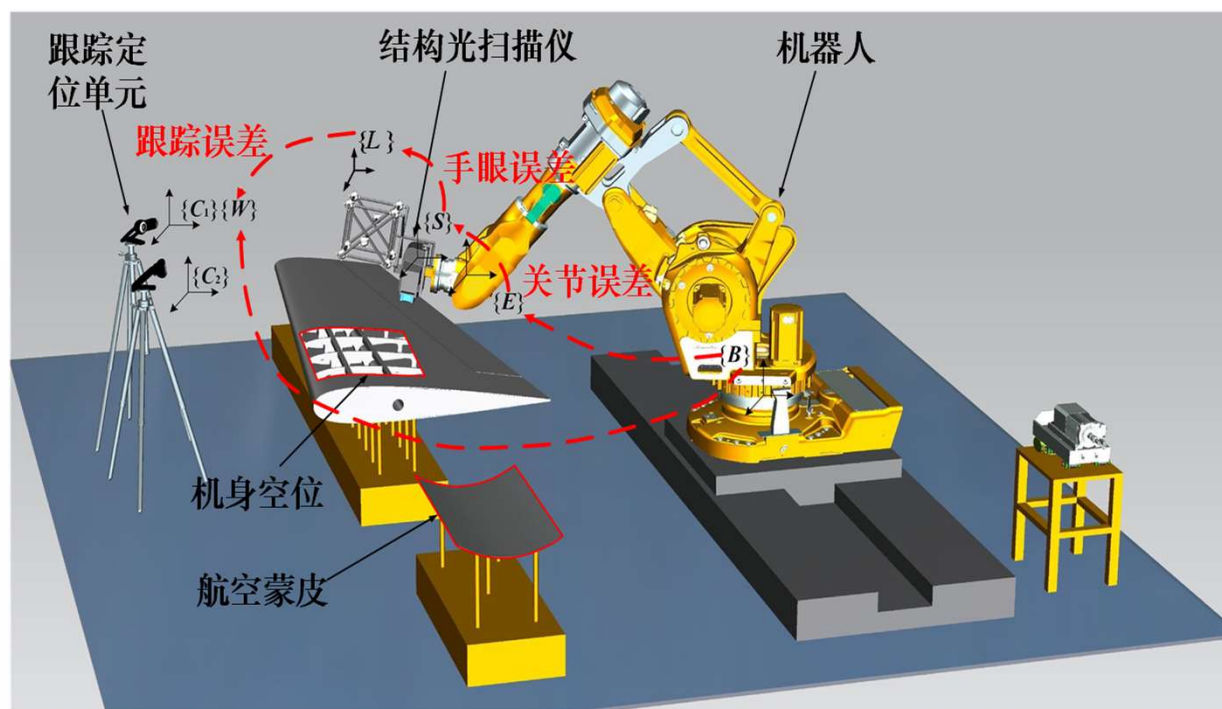
人工智能系

晁飞副教授



位姿描述—机器人铣削修边应用场景

飞机蒙皮加工系统：机器人、扫描仪、跟踪仪、工具、工件



位姿描述—机器人操作（磨抛）应用场景

大叶片加工工艺：模锻——铣削——变形矫正——磨抛



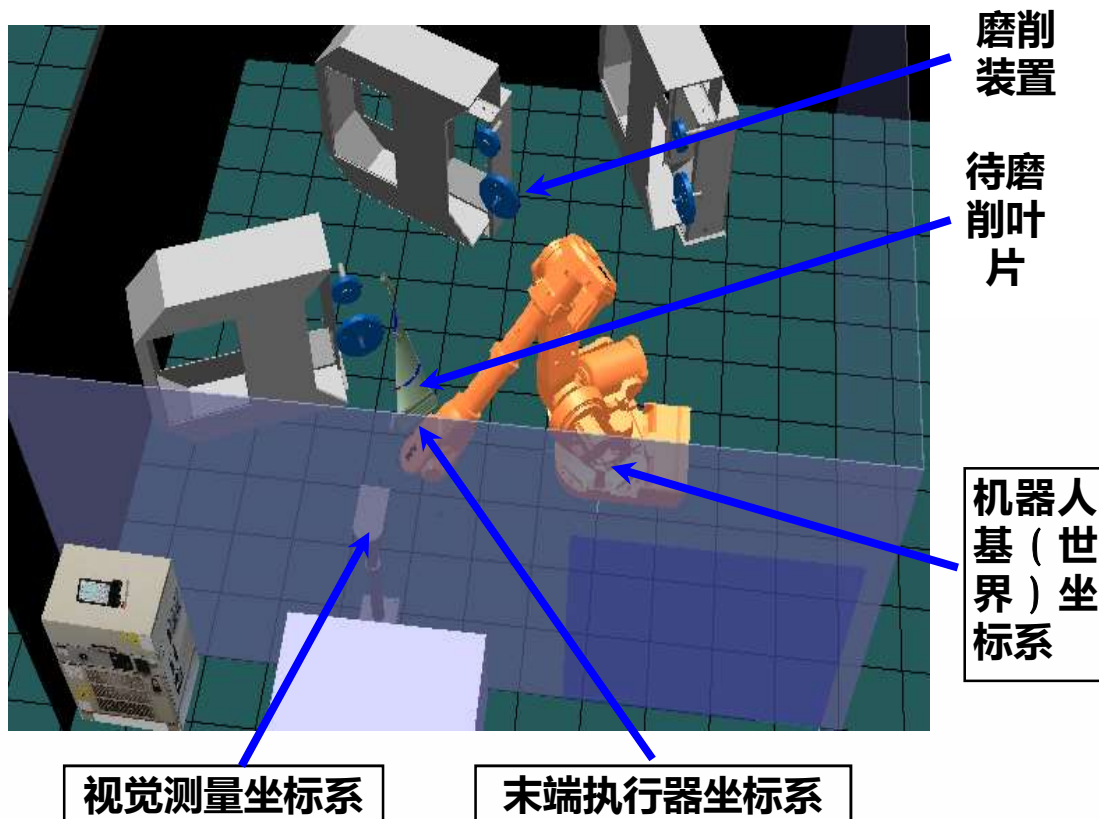
手工磨抛



机器人磨抛

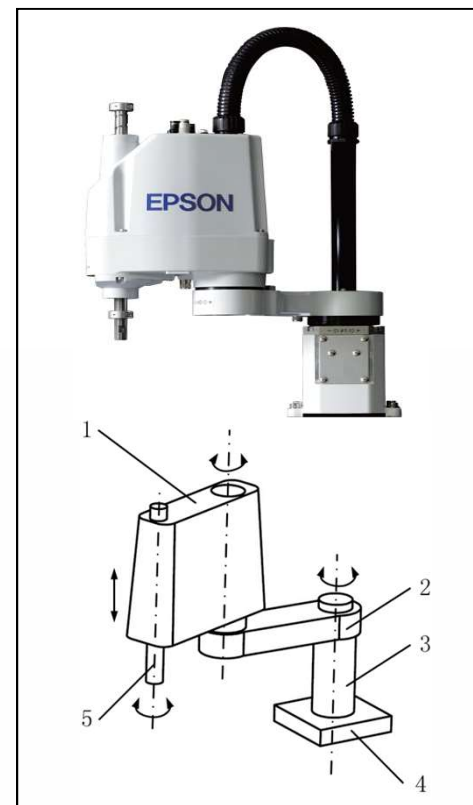
位姿描述—机器人操作（磨抛）应用场景

机器人磨抛涉及的坐标系：



本章提纲

- 3.1 刚体位姿描述
- 3.2 齐次坐标和齐次变换
- 3.3 运动算子
- 3.4 变换矩阵的运算
- 3.5 欧拉角与RPY角
- 3.6 旋转变换通式
- 3.7 位姿的综合
- 3.8 计算的复杂性



3.1 刚体位姿描述

- 研究机器人操作臂的运动：涉及各连杆位姿关系、连杆与周围环境（操作对象和障碍物）的关系。我们把操作臂的**各个连杆、操作对象、工具、工件和障碍物**都当成刚体。
- 刚体位姿描述：**齐次变换（矩阵）、矢量法、四元数**

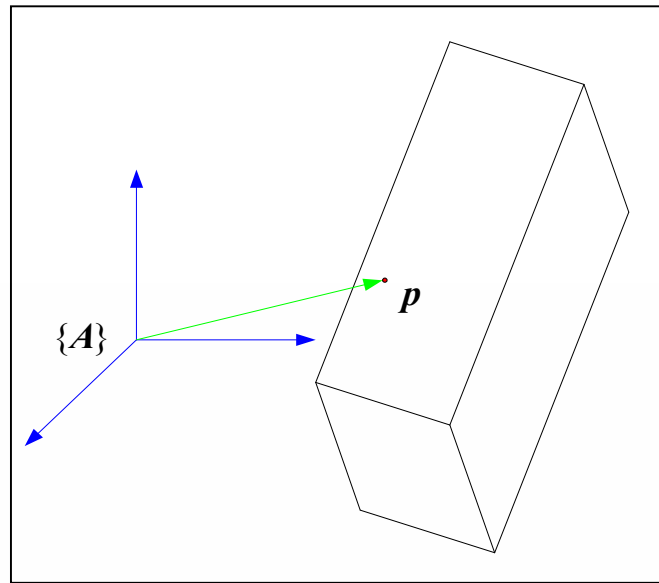
- ◆ 位姿 (Location) = 位置 (Position) + 姿态 (Orientation)
- ◆ 齐次变换法
 - 将运动、变换和映射与矩阵运算联系起来；
 - 在操作臂运动/动力学、机器人控制算法、计算机图学、视觉信息处理、手-眼建模标定都有广泛应用。
- ◆ 描述刚体在坐标系中的相对位姿（描述）； Description
- ◆ 表示刚体运动前后位姿描述的变换（算子）； Operator
- ◆ 把点从一个坐标系映射到另一个坐标系（映射）； Mapping

3.1 位置的描述（位置矢量）

□ 在坐标系 $\{A\}$ 中，空间任意一点可表示为列矢量 ${}^A p$

$${}^A \mathbf{p} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$

直角坐标系



□ 符号表示：矢量/向量等（小写粗体）、矩阵/坐标系/群等（大写粗体）、变量/角度等（小写非斜体）

3.1 旋转矩阵与旋转群

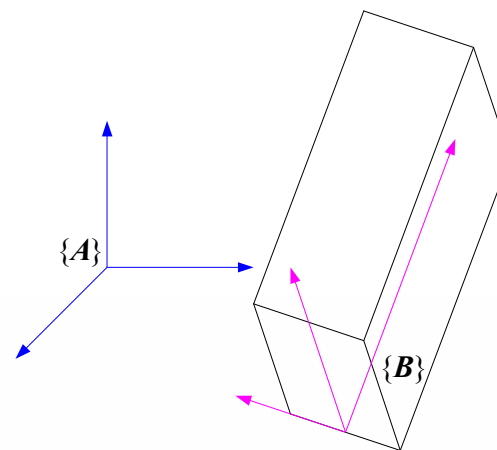
□ 坐标系 $\{B\}$ 相对于坐标系 $\{A\}$ 的旋转矩阵

$${}^A_B R = \begin{bmatrix} {}^A x_B & {}^A y_B & {}^A z_B \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

${}^A_B R$ 是正交矩阵, 满足如下关系:

$${}^A x_B \cdot {}^A x_B = {}^A y_B \cdot {}^A y_B = {}^A z_B \cdot {}^A z_B = 1$$

$${}^A x_B \cdot {}^A y_B = {}^A y_B \cdot {}^A z_B = {}^A z_B \cdot {}^A x_B = 0$$



□ 所以旋转矩阵中9个元素只有3个独立变量

□ 还满足: ${}^A_B R^{-1} = {}^A_B R^T$; $\det({}^A_B R) = +1$

上标T表示转置, det表示行列式符号

刚体-1

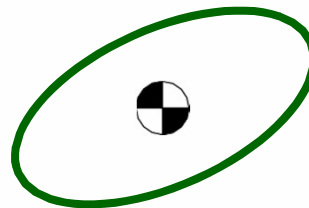
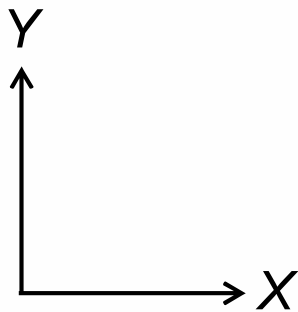
□ 一个刚体(Rigid body)的状态该如何描述？

◆ 平面：

移动？ DOFs、转动？ DOF

Degree of freedom

$\{W\}$ world frame



刚体-1

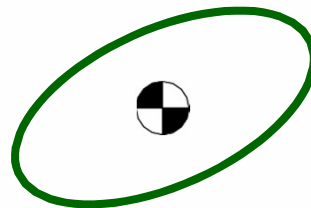
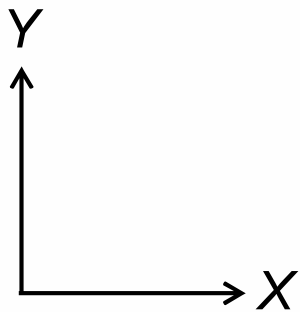
□ 一个刚体(Rigid body)的状态该如何描述?

◆ 平面:

移动 2 DOFs、转动 1 DOF

Degree of freedom

$\{W\}$ world frame



刚体 -2

□ 一个刚体(Rigid body)的状态该如何描述？

◆ 平面：

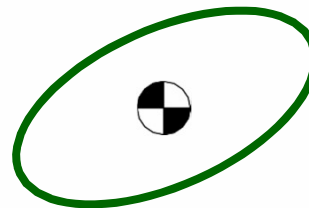
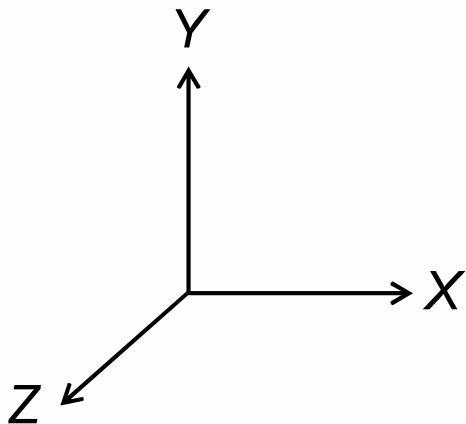
移动 2 DOFs、转动 1 DOF

Degree of freedom

◆ 空间：

移动 ? DOFs、转动 ? DOFs

$\{W\}$ world frame



刚体 -2

□ 一个刚体(Rigid body)的状态该如何描述？

◆ 平面：

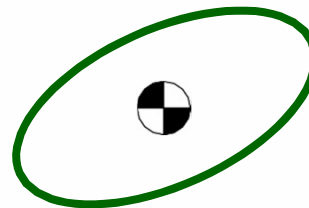
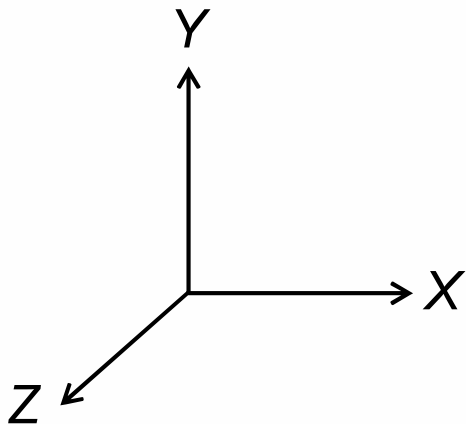
移动 2 DOFs、转动 1 DOF

Degree of freedom

◆ 空间：

移动 3 DOFs、转动 3 DOFs

$\{W\}$ world frame

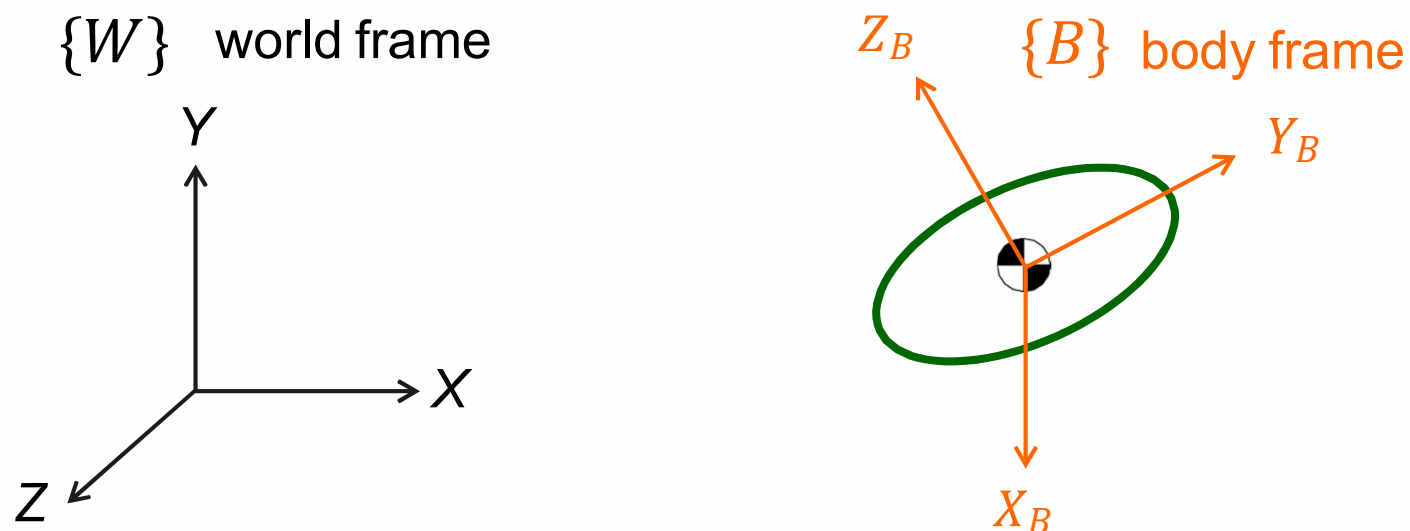


刚体-3

□ 该如何整合表达刚体的状态？

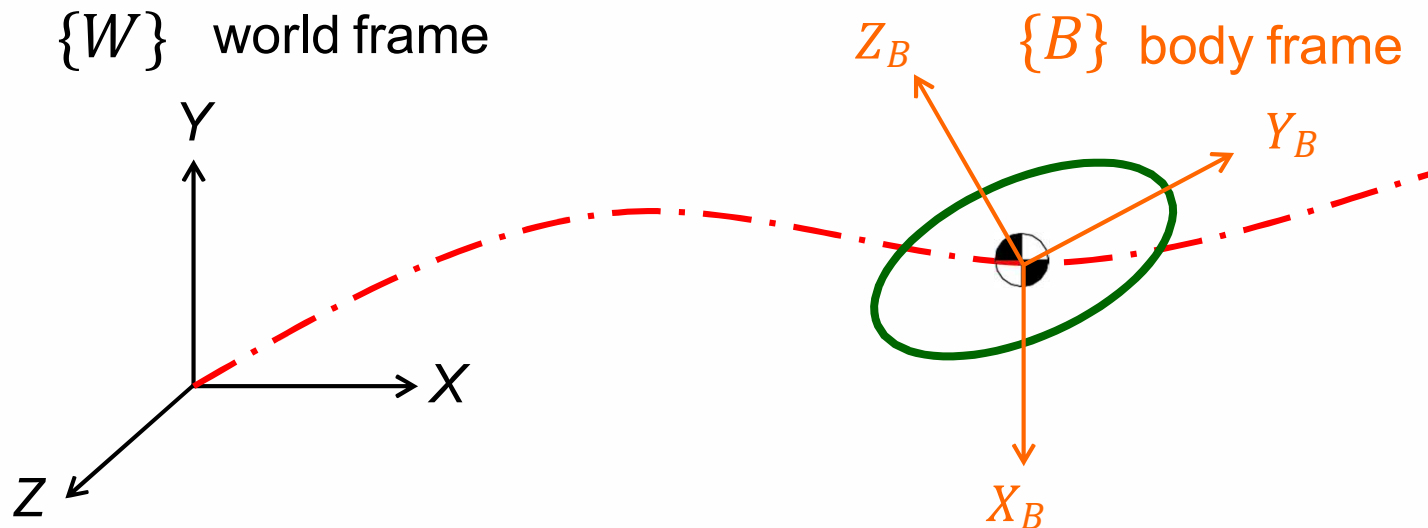
⇒ 在刚体(Rigid body)上建立frame，常建立在质心上

- ◆ 移动：由body frame 的原点位置判定
- ◆ 转动：由body frame的姿态判定



导读 -4

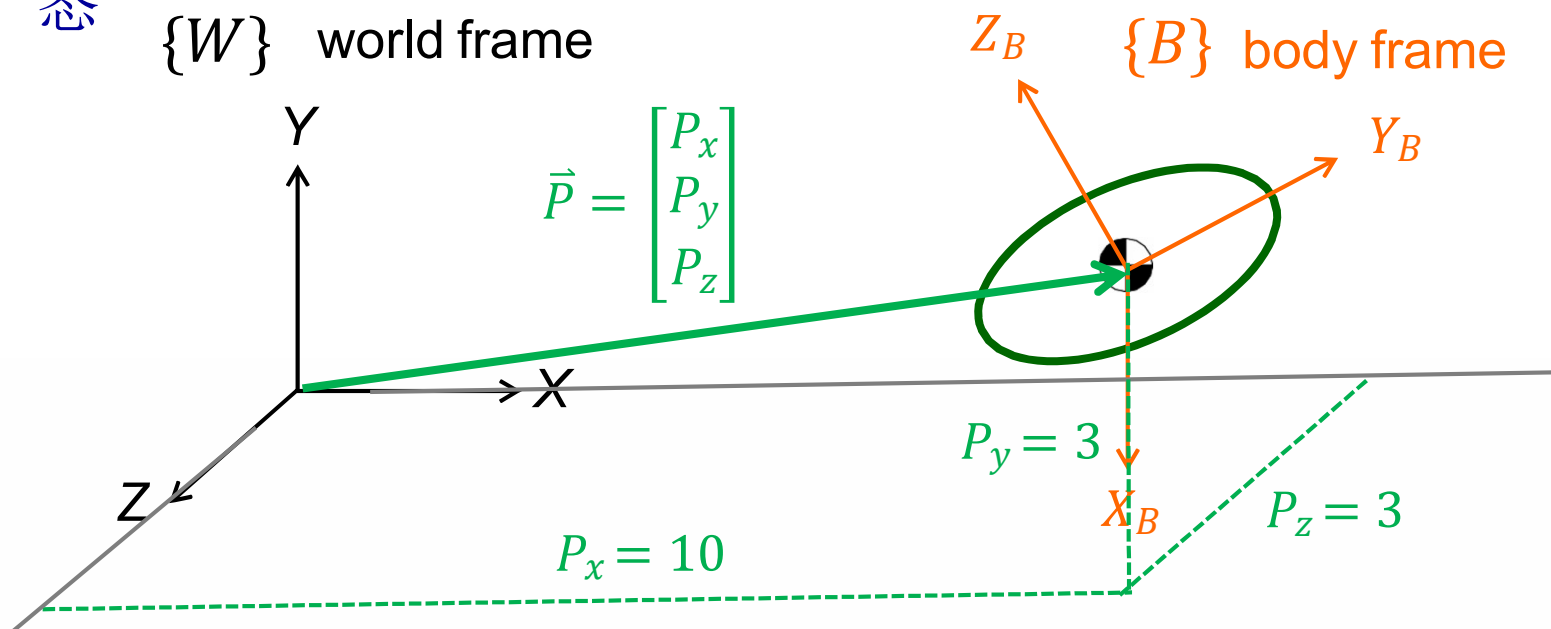
- 一个刚体(Rigid body)的「运动」状态该如何描述?
 - ◆ 利用各个DOF的微分，将位移和姿态 (displacement / orientation) 转换到速度 (velocity) 和加速度 (acceleration) 等运动状态



移动 -1

- 移动：以向量（vector） $\vec{P}$ 来描述 $\{B\}$ 的原点相对于 $\{A\}$ 的状态

态



- Ex: $\vec{P} = \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \begin{bmatrix} 10 \\ 3 \\ 3 \end{bmatrix}$

移动 -2

□ 向量可表达空间关系的两个方式

- ◆ A position in space (i.e., position vector)

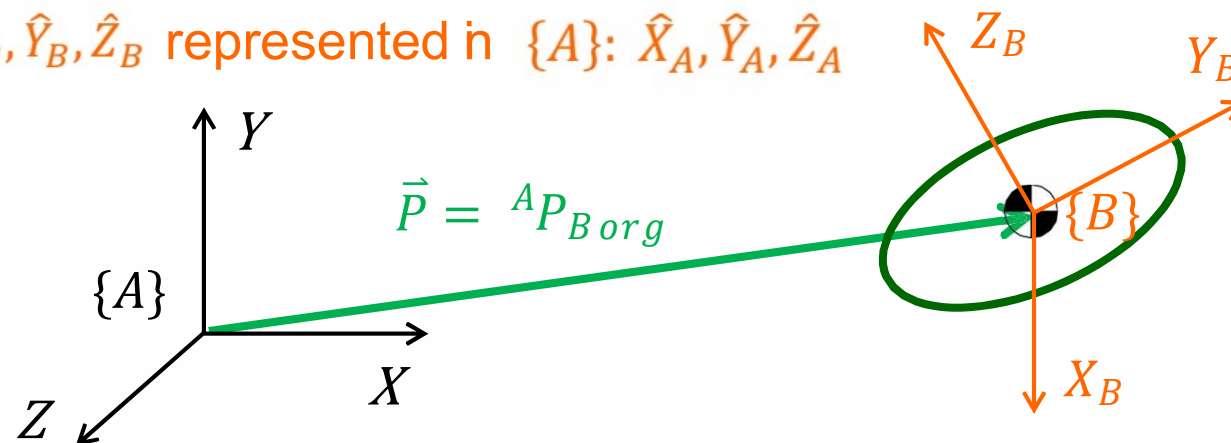
如同前一页内容，以此方式描述body frame原点

$$\vec{P} = \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = {}^A P_{B \text{ org}} = \text{origin of } \{B\} \text{ represented in } \{A\}$$

- ◆ A vector (i.e., displacement, frame basis)

以此方式表达body frame上principal axes的方向

$\{B\}$: $\hat{X}_B, \hat{Y}_B, \hat{Z}_B$ represented in $\{A\}$: $\hat{X}_A, \hat{Y}_A, \hat{Z}_A$

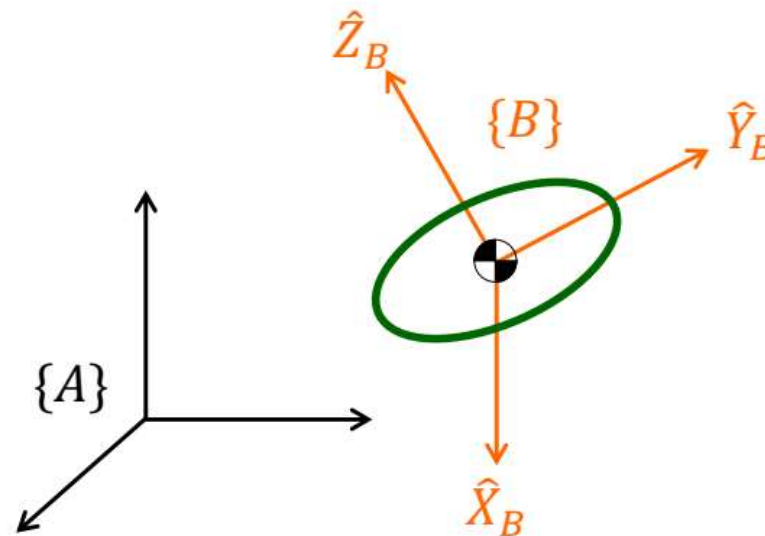


转动 -1

- 转动：描述{B}相对于{A}之姿态---Rotation Matrix

$$\begin{aligned} \overset{\substack{\uparrow \\ \text{B relative} \\ \text{to A}}}{A_B R} &= \begin{bmatrix} | & | & | \\ {}^A \hat{X}_B & {}^A \hat{Y}_B & {}^A \hat{Z}_B \\ | & | & | \end{bmatrix} \\ &\quad \text{"column vector"} \\ &= \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} \\ &\quad \text{"direct cosines"} \end{aligned}$$

R的三个columns即为frame {B}的basis: $\hat{X}_B, \hat{Y}_B, \hat{Z}_B$ (由{A}看)



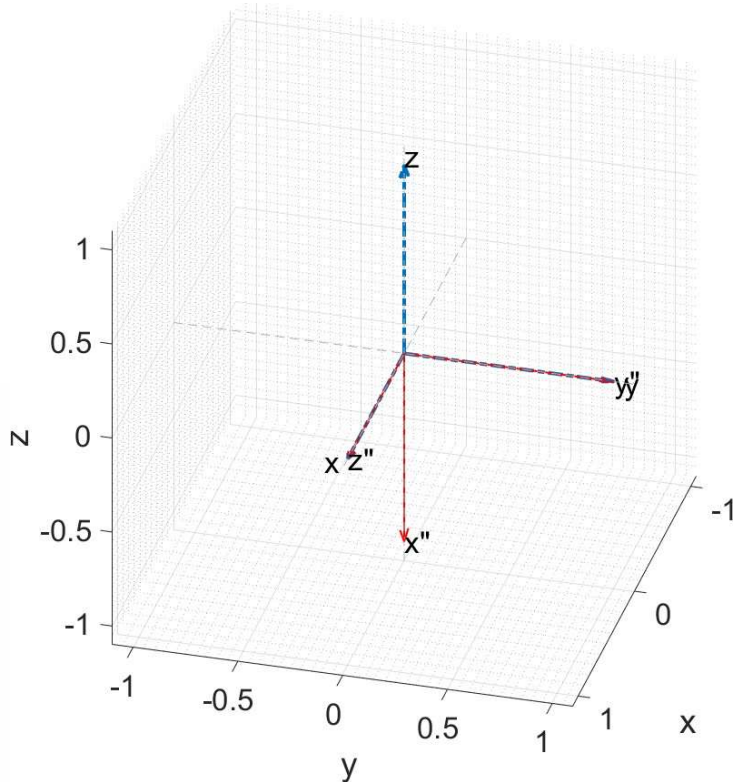
转动 -2

$$= \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}$$

□ Ex: $\{B\}$ 相对于 $\{A\}$ 之姿态 ${}^A_B R = ?$

蓝虚线: World Frame $\{A\}$

红实线: Body Frame $\{B\}$



$$\{B\} \text{ 的 } x'' \text{ 轴为 } \{A\} \text{ 的 } z \text{ 轴反向} \Rightarrow {}^A \hat{X}_B = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

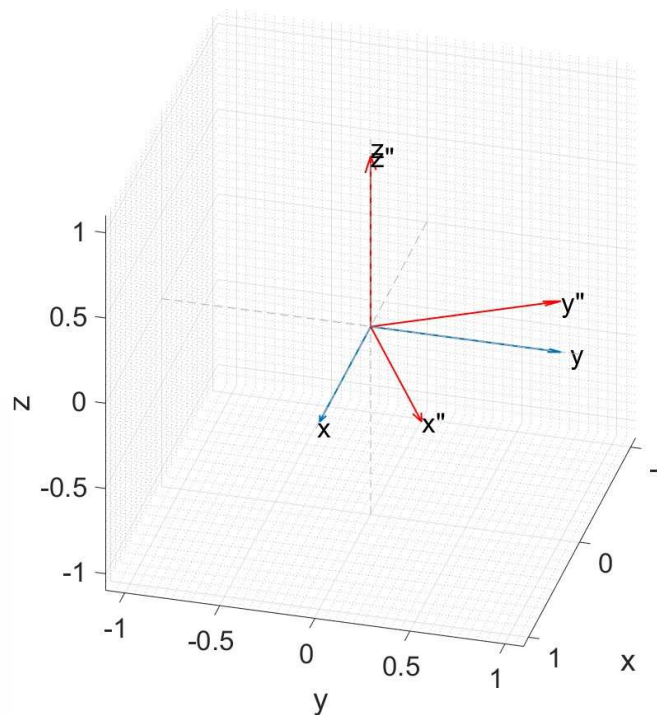
$$\{B\} \text{ 的 } y'' \text{ 轴与 } \{A\} \text{ 的 } y \text{ 轴重迭} \Rightarrow {}^A \hat{Y}_B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\{B\} \text{ 的 } z'' \text{ 轴与 } \{A\} \text{ 的 } x \text{ 轴重迭} \Rightarrow {}^A \hat{Z}_B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{因此, } \{B\} \text{ 相对于 } \{A\} \text{ 之姿态: } {}^A_B R = \begin{bmatrix} | & | & | \\ {}^A \hat{X}_B & {}^A \hat{Y}_B & {}^A \hat{Z}_B \\ | & | & | \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

转动 -3

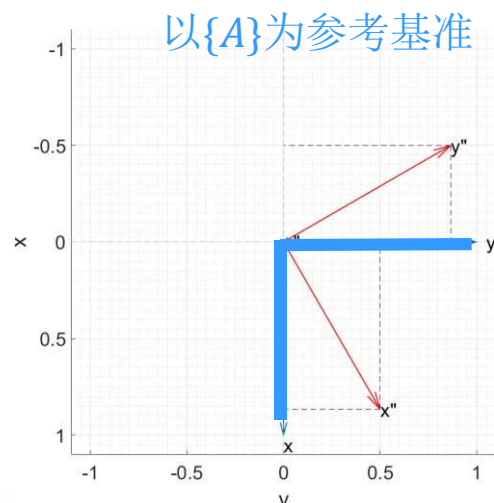
□ Ex: $\{B\}$ 相对于 $\{A\}$ 之姿态 ${}^A_B R = ?$



蓝虚线: World Frame $\{A\}$

红实线: Body Frame $\{B\}$

上视图



$${}^A \hat{X}_B = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} 0.866 \\ 0.5 \\ 0 \end{bmatrix}$$

$${}^A \hat{Y}_B = \begin{bmatrix} \hat{Y}_B \cdot \hat{X}_A \\ \hat{Y}_B \cdot \hat{Y}_A \\ \hat{Y}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} -0.5 \\ 0.866 \\ 0 \end{bmatrix}$$

$${}^A \hat{Z}_B = \begin{bmatrix} \hat{Z}_B \cdot \hat{X}_A \\ \hat{Z}_B \cdot \hat{Y}_A \\ \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$\{B\}$ 相对于 $\{A\}$ 之姿态:

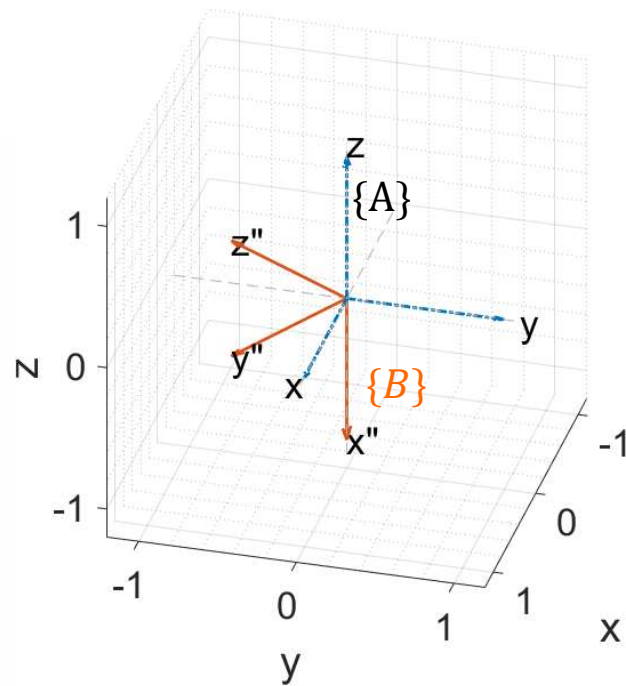
$${}^A_B R = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

转动 -4

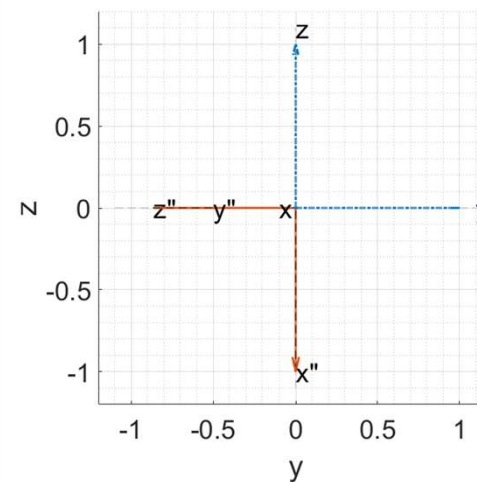
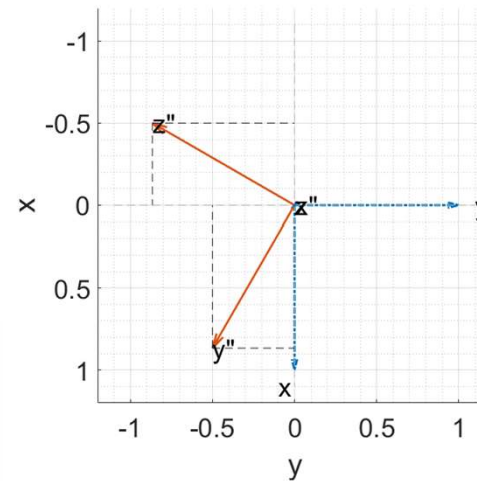
□ 课堂练习: $\{B\}$ 相对于 $\{A\}$ 的姿态 ${}^A_B R = ?$

蓝虚线: World Frame $\{A\}$

红实线: Body Frame $\{B\}$



投影到
XY/YZ平面

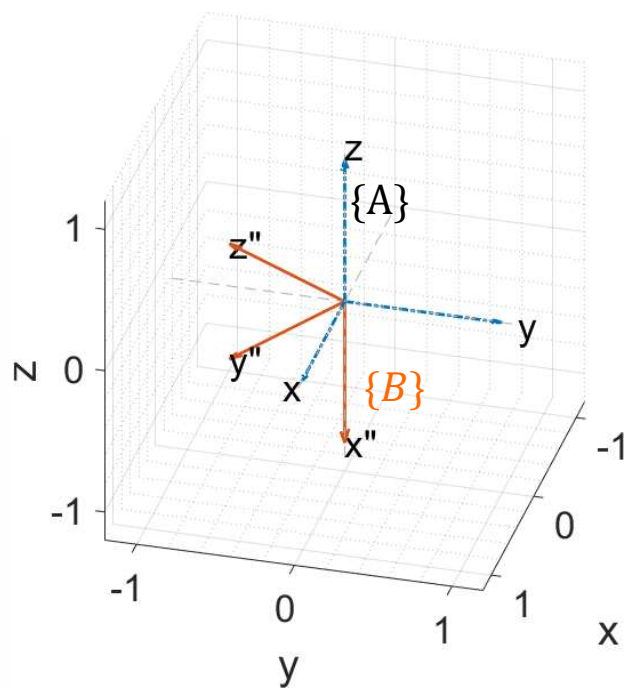


转动 -4

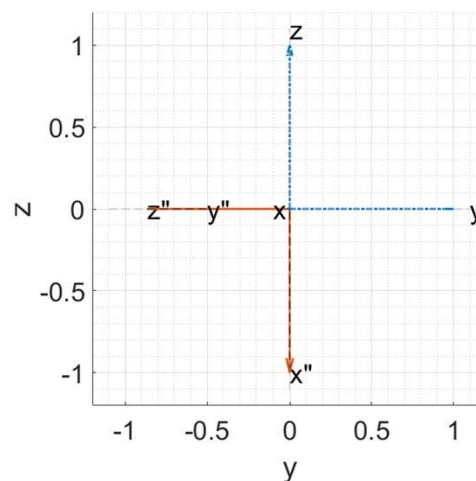
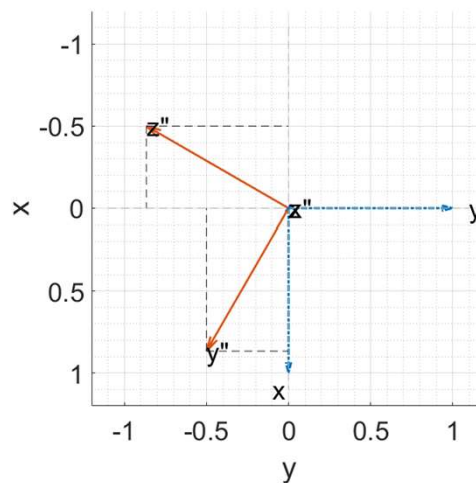
□ 课堂练习: $\{B\}$ 相对于 $\{A\}$ 的姿态 ${}^A_B R = ?$

蓝虚线: World Frame $\{A\}$

红实线: Body Frame $\{B\}$



投影到
XY/YZ平面



$${}^A\hat{X}_B = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

$${}^A\hat{Y}_B = \begin{bmatrix} \hat{Y}_B \cdot \hat{X}_A \\ \hat{Y}_B \cdot \hat{Y}_A \\ \hat{Y}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} 0.866 \\ -0.5 \\ 0 \end{bmatrix}$$

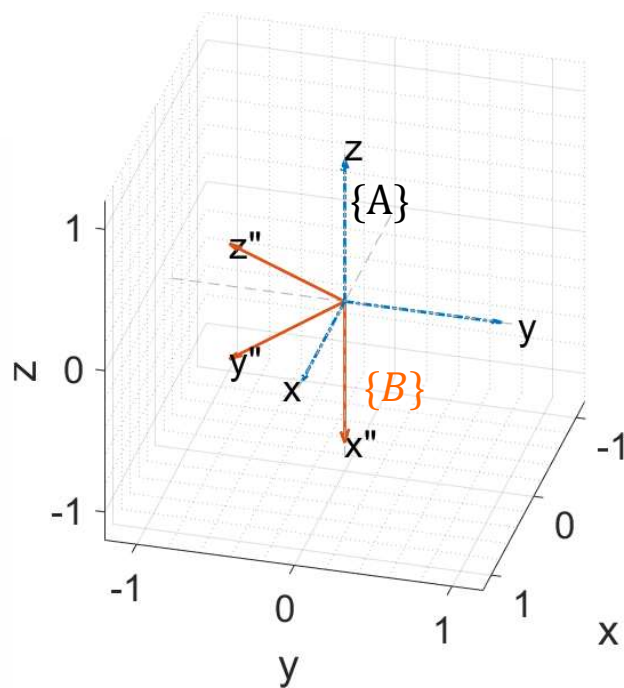
$${}^A\hat{Z}_B = \begin{bmatrix} \hat{Z}_B \cdot \hat{X}_A \\ \hat{Z}_B \cdot \hat{Y}_A \\ \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} -0.5 \\ -0.866 \\ 0 \end{bmatrix}$$

转动 -4

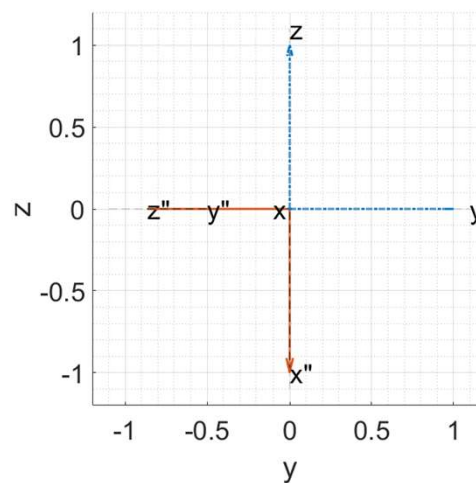
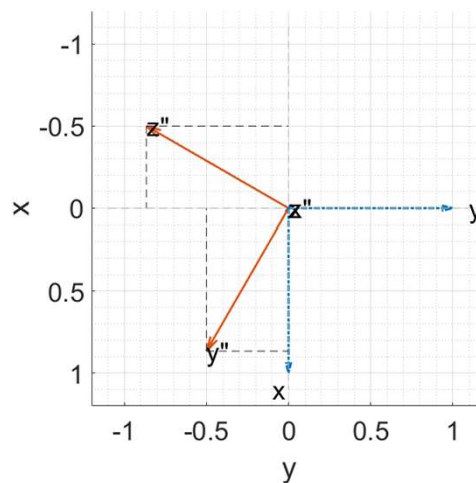
□ 课题联系: $\{B\}$ 相对于 $\{A\}$ 之姿态 ${}^A R_B = ?$

蓝虚线: World Frame $\{A\}$

红实线: Body Frame $\{B\}$



投影到
XY/YZ平面



A.
$$\begin{bmatrix} 0 & -0.866 & 0.5 \\ 0 & 0.5 & 0.866 \\ -1 & 0 & 0 \end{bmatrix}$$

B.
$$\begin{bmatrix} 0 & 0.866 & -0.5 \\ 0 & -0.5 & -0.866 \\ -1 & 0 & 0 \end{bmatrix}$$

C.
$$\begin{bmatrix} 0 & 0 & -1 \\ 0.866 & -0.5 & 0 \\ -0.5 & -0.866 & 0 \end{bmatrix}$$

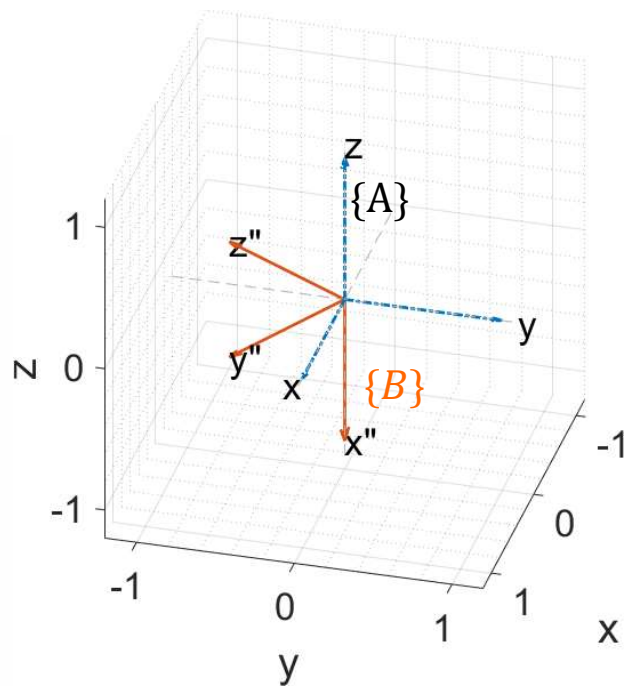
D.
$$\begin{bmatrix} 0 & 0 & -1 \\ -0.866 & 0.5 & 0 \\ 0.5 & 0.866 & 0 \end{bmatrix}$$

转动 -4

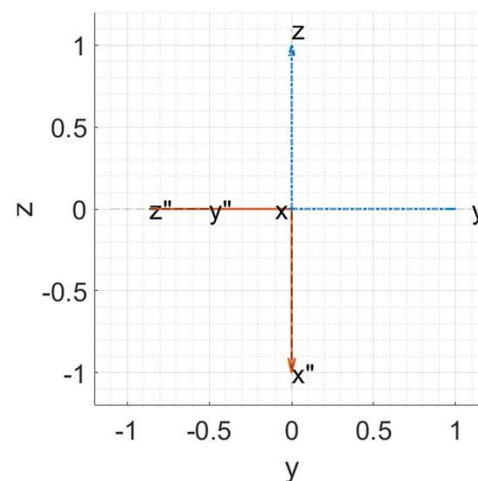
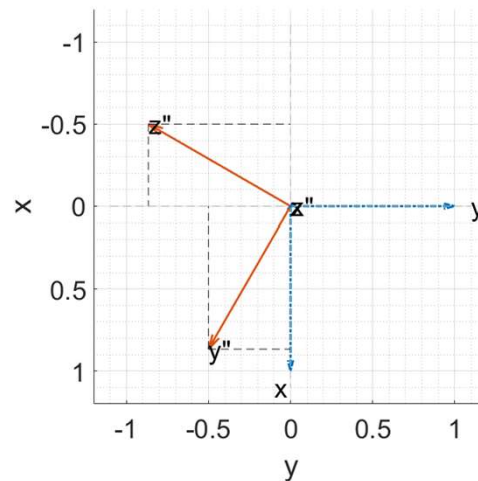
□ In-video Quiz: $\{B\}$ 相对于 $\{A\}$ 之姿态 ${}^A_B R = ?$

蓝虚线: World Frame $\{A\}$

红实线: Body Frame $\{B\}$



投影到
XY/YZ平面



A.
$$\begin{bmatrix} 0 & -0.866 & 0.5 \\ 0 & 0.5 & 0.866 \\ -1 & 0 & 0 \end{bmatrix}$$

B.
$$\begin{bmatrix} 0 & 0.866 & -0.5 \\ 0 & -0.5 & -0.866 \\ -1 & 0 & 0 \end{bmatrix}$$

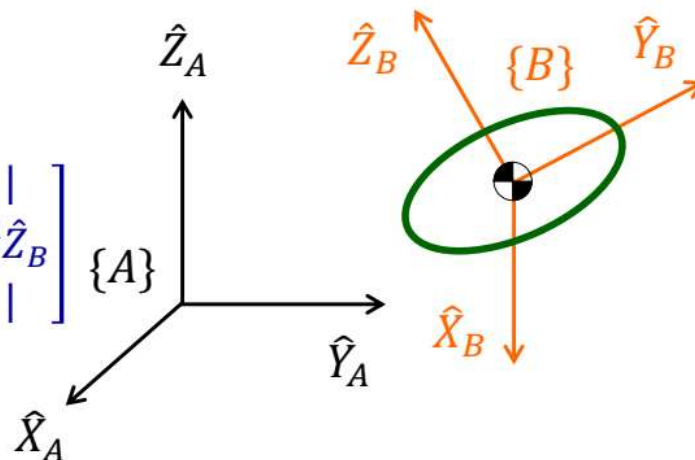
C.
$$\begin{bmatrix} 0 & 0 & -1 \\ 0.866 & -0.5 & 0 \\ -0.5 & -0.866 & 0 \end{bmatrix}$$

D.
$$\begin{bmatrix} 0 & 0 & -1 \\ -0.866 & 0.5 & 0 \\ 0.5 & 0.866 & 0 \end{bmatrix}$$

Rotation Matrix -1

□ 特性一

$${}^A_B R = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix} \{A\}$$



前后向量互换

$$= \begin{bmatrix} \hat{X}_A \cdot \hat{X}_B & \hat{X}_A \cdot \hat{Y}_B & \hat{X}_A \cdot \hat{Z}_B \\ \hat{Y}_A \cdot \hat{X}_B & \hat{Y}_A \cdot \hat{Y}_B & \hat{Y}_A \cdot \hat{Z}_B \\ \hat{Z}_A \cdot \hat{X}_B & \hat{Z}_A \cdot \hat{Y}_B & \hat{Z}_A \cdot \hat{Z}_B \end{bmatrix} = \begin{bmatrix} - & {}^B\hat{X}_A^T & - \\ - & {}^B\hat{Y}_A^T & - \\ - & {}^B\hat{Z}_A^T & - \end{bmatrix}$$

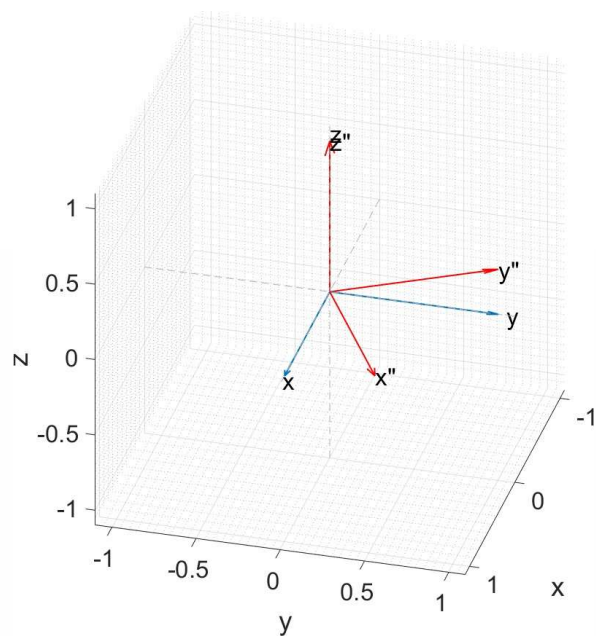
$$= \begin{bmatrix} | & | & | \\ {}^B\hat{X}_A & {}^B\hat{Y}_A & {}^B\hat{Z}_A \\ | & | & | \end{bmatrix}^T = {}^B_A R^T$$

⇒ ${}^A_B R = {}^B_A R^T$

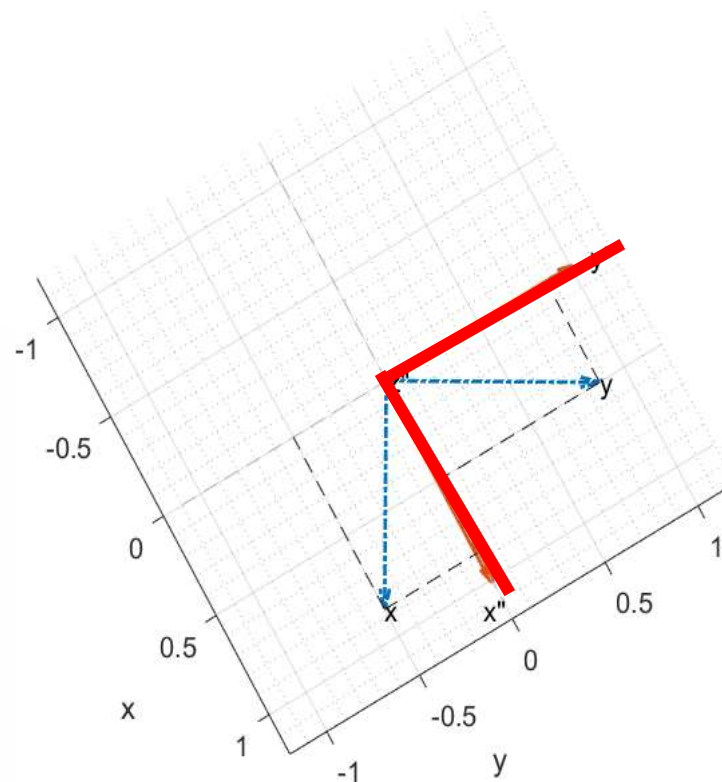
Rotation Matrix -2

□ Ex: $\{A\}$ 相对于 $\{B\}$ 之姿态 ${}^B_A R = ?$

以 $\{B\}$ 为参考基准



俯视图

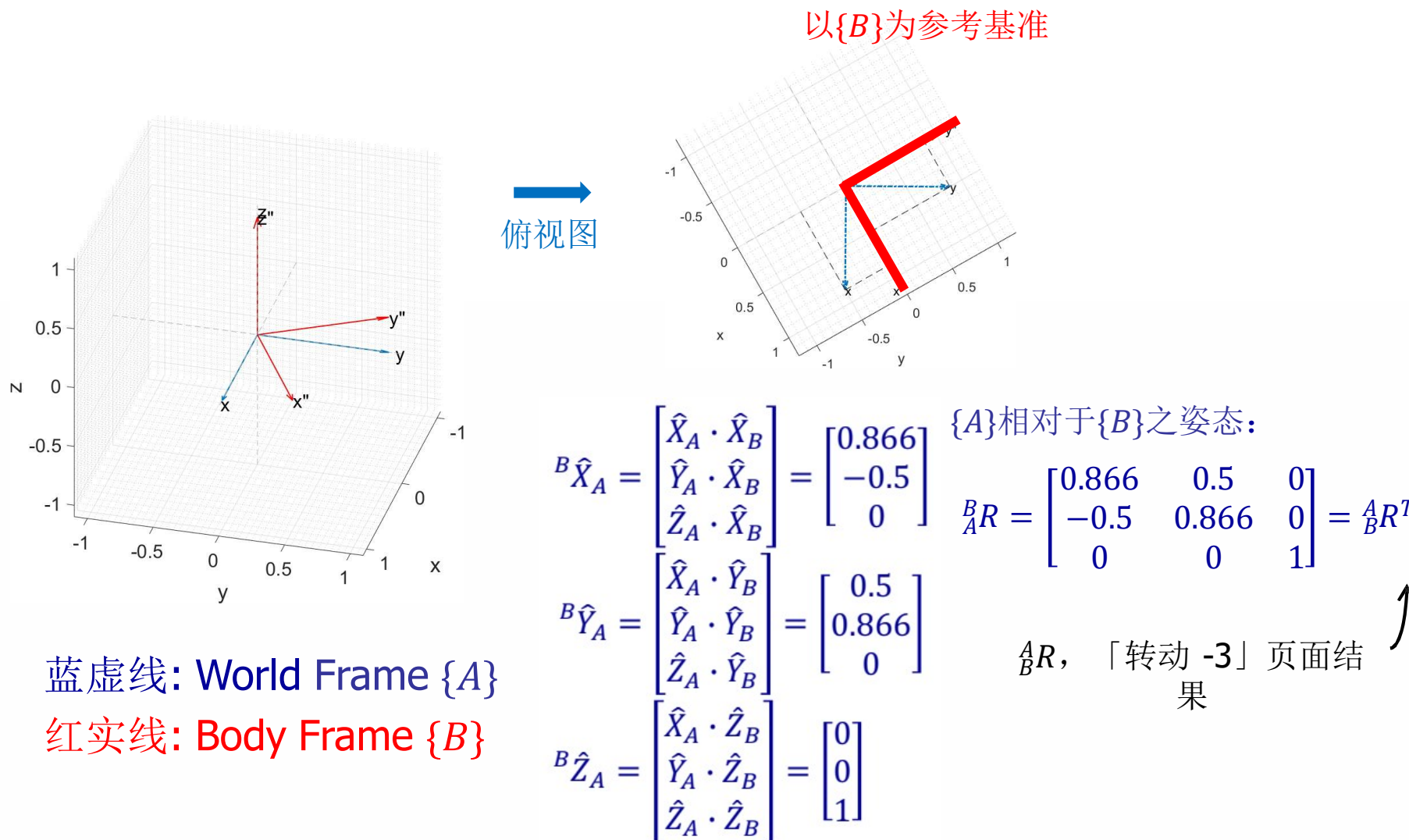


蓝虚线: World Frame $\{A\}$

红实线: Body Frame $\{B\}$

Rotation Matrix -2

□ Ex: {A}相对于{B}之姿态 ${}^B_A R = ?$



Rotation Matrix -3

□ 特性二

$${}^A_B R^T {}^A_B R = \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix}^T \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix} \{A\}$$

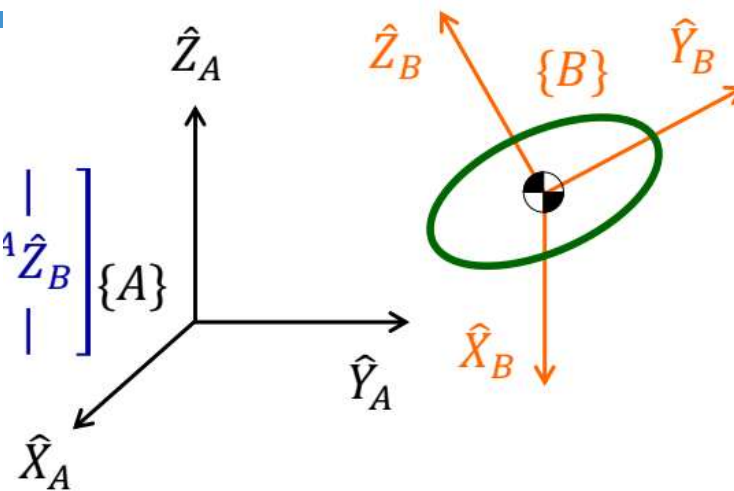
$$= \begin{bmatrix} - & {}^A\hat{X}_B^T & - \\ - & {}^A\hat{Y}_B^T & - \\ - & {}^A\hat{Z}_B^T & - \end{bmatrix} \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix}$$

$$= I_3$$

↖ 3x3 identity matrix

$$= {}^A_B R^{-1} {}^A_B R$$

$$\Rightarrow {}^A_B R^T = {}^A_B R^{-1} = {}^B_A R$$



Rotation Matrix -4

□ A 3x3 正交矩阵 Q

$$QQ^T = Q^TQ = I$$

- ◆ 总是可逆的 $Q^{-1} = Q^T$
- ◆ 列：正交基
 - ◆ 长度=1
 - ◆ 相互垂直
- ◆ Rotation matrix (R)有9个数字，但上列两个条件置入了6个 constraints，所以R只有3个DOFs，与空间中转动具有3 DOFs相符
- ◆ Determinant =1 (rotation); =-1 (reflection)

Rotation Matrix -5 - 第二个功能

- Rotation matrix 除描述{B}相对于{A}之姿态，也可用于转换向量之坐标

original coordinate ${}^B P = {}^B P_x \hat{X}_B + {}^B P_y \hat{Y}_B + {}^B P_z \hat{Z}_B$

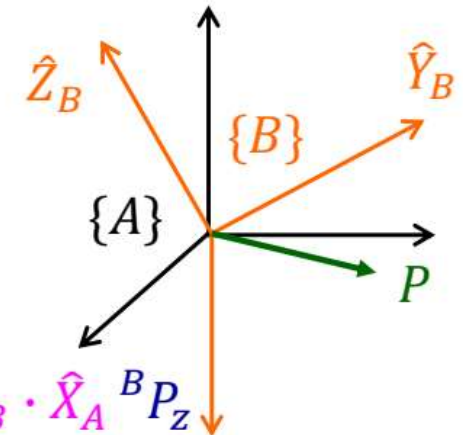
new coordinate ${}^A P = {}^A P_x \hat{X}_A + {}^A P_y \hat{Y}_A + {}^A P_z \hat{Z}_A$

where ${}^A P_x = {}^B P \cdot \hat{X}_A = \hat{X}_B \cdot \hat{X}_A {}^B P_x + \hat{Y}_B \cdot \hat{X}_A {}^B P_y + \hat{Z}_B \cdot \hat{X}_A {}^B P_z$

${}^A P_y = {}^B P \cdot \hat{Y}_A = \hat{X}_B \cdot \hat{Y}_A {}^B P_x + \hat{Y}_B \cdot \hat{Y}_A {}^B P_y + \hat{Z}_B \cdot \hat{Y}_A {}^B P_z$

${}^A P_z = {}^B P \cdot \hat{Z}_A = \hat{X}_B \cdot \hat{Z}_A {}^B P_x + \hat{Y}_B \cdot \hat{Z}_A {}^B P_y + \hat{Z}_B \cdot \hat{Z}_A {}^B P_z$

$$\Rightarrow {}^A P = \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} {}^B \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = {}^A_B R {}^B P$$

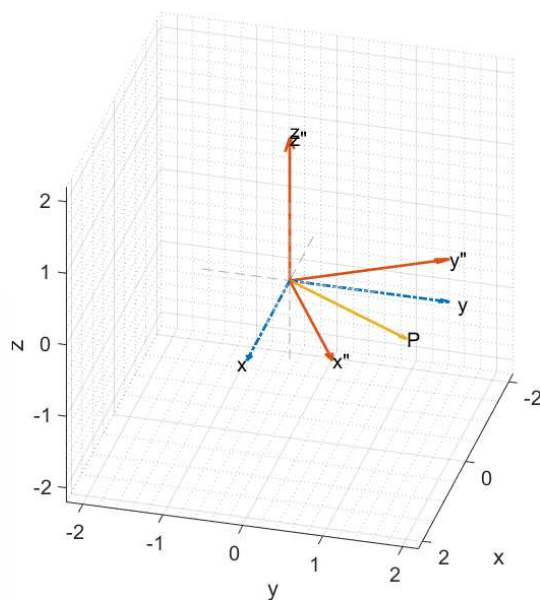


和「转动-1」页matrix相同，为rotation matrix

Rotation Matrix -6

□ Ex: 若 $\{B\}$ 和 $\{A\}$ 的相对状态同「转动 -3」 页面所示, 假设

$${}^B P = [1.732 \quad 1 \quad 0]^T, \quad {}^A P = ?$$



$${}^A_B R = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

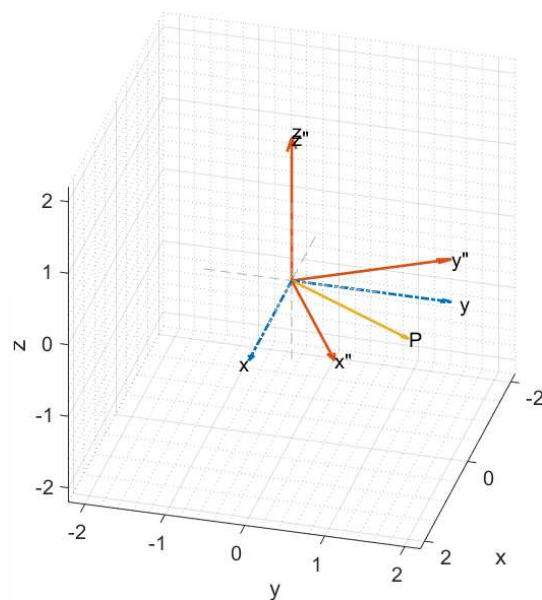
蓝虚线: World Frame $\{A\}$

红实线: Body Frame $\{B\}$

Rotation Matrix -6

□ Ex: 若{B}和{A}的相对状态同「转动 -3」页面所示, 假设

$${}^B P = [1.732 \quad 1 \quad 0]^T, \quad {}^A P = ?$$



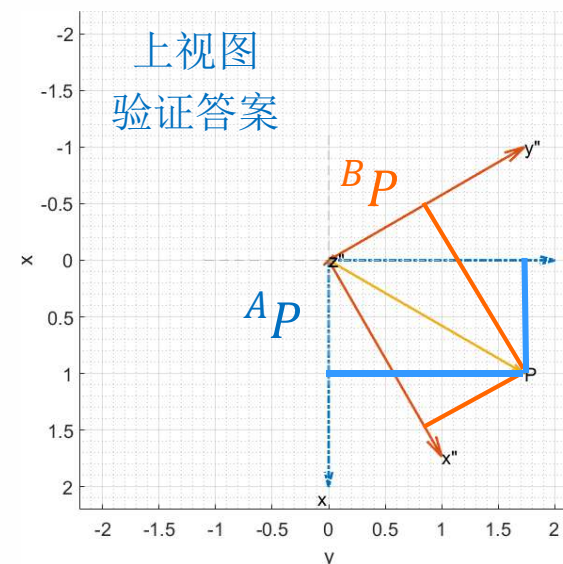
$${}^A P = {}^A_B R {}^B P$$

$${}^A P = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1.732 \\ 1 \\ 0 \end{bmatrix}$$

$${}^A P = \begin{bmatrix} 1 \\ 1.732 \\ 0 \end{bmatrix}$$

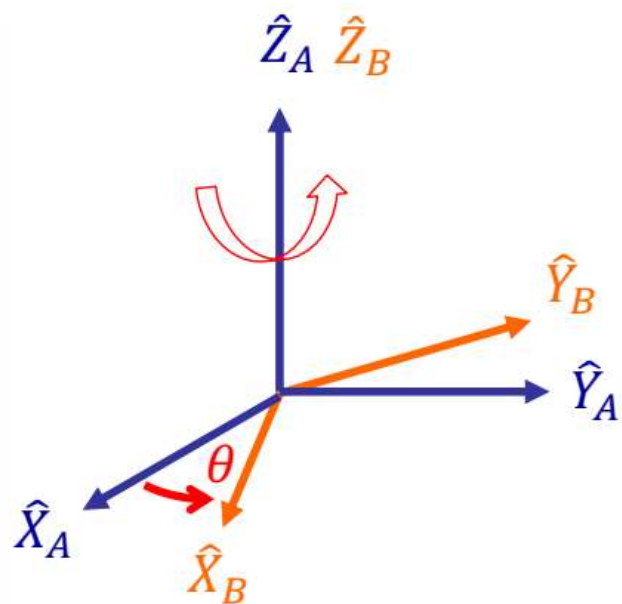
蓝虚线: World Frame {A}

红实线: Body Frame {B}



Rotation Matrix -7

- Rotation matrix的第三个功能，可进一步来描述物体转动的状态
- 以对三个principal axes旋转的matrix为基础
- About $\hat{Z}_A$ with θ



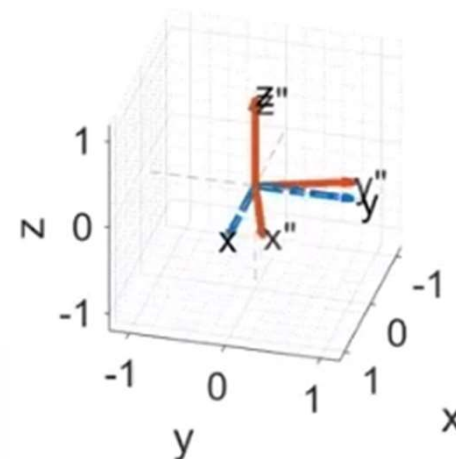
旋转角度

$$R_{\hat{Z}_A}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

旋转轴

$$= \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Note: ${}^A_B R = \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix}$



Rotation Matrix -8

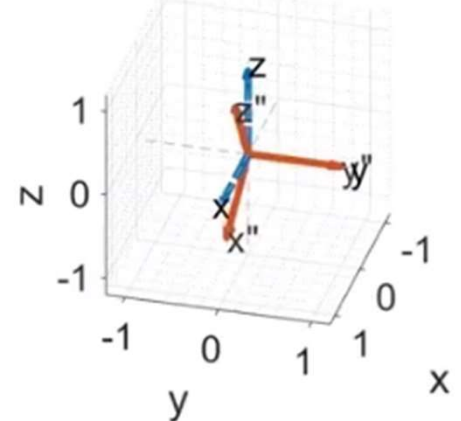
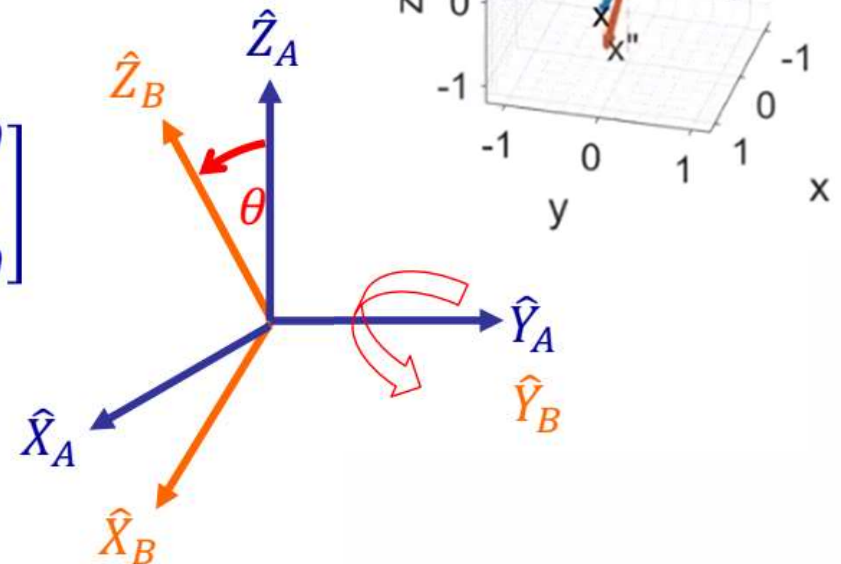
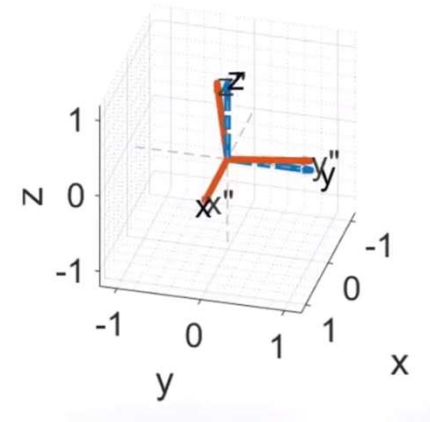
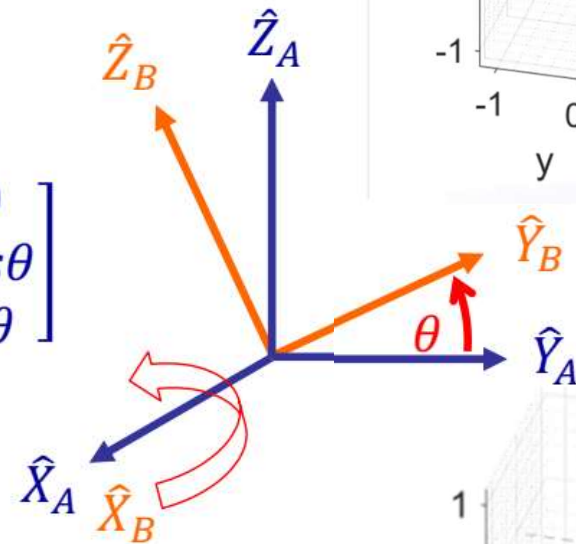
- About $\hat{X}_A$ with θ

$$R_{\hat{X}_A}(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\theta & -s\theta \\ 0 & s\theta & c\theta \end{bmatrix}$$

- About $\hat{Y}_A$ with θ

$$R_{\hat{Y}_A}(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} = \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix}$$

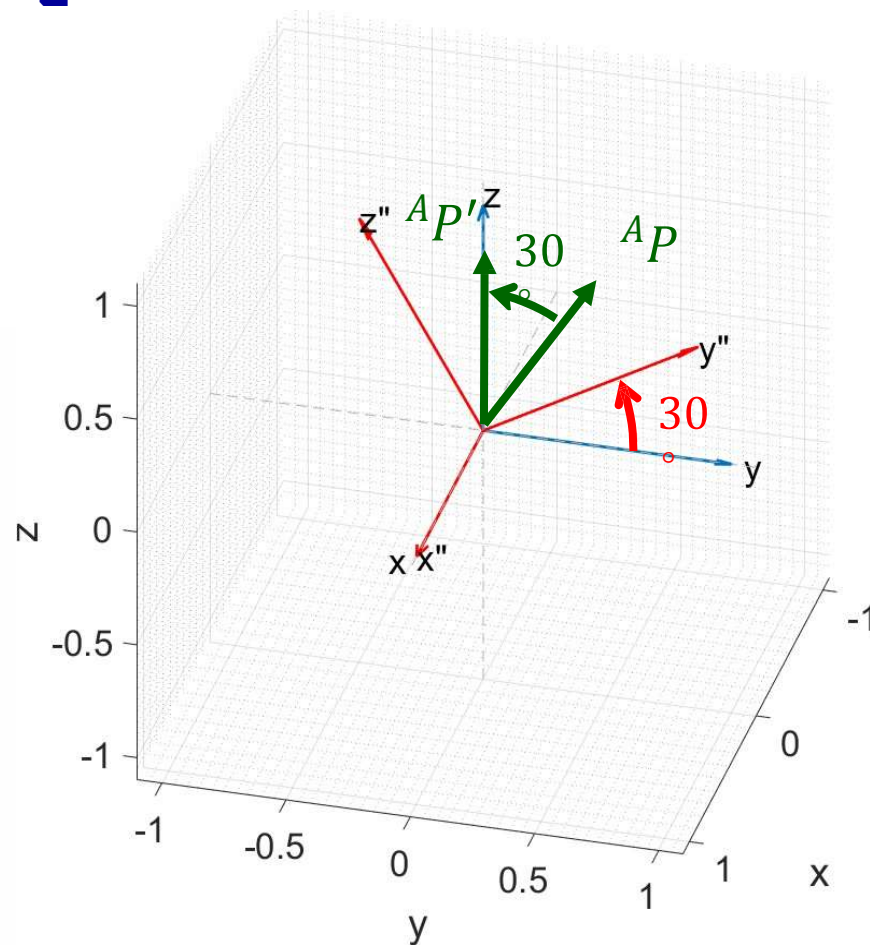
Note: ${}^A_B R = \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix}$



Rotation Matrix -9

□ Ex: ${}^A P = [0 \quad 1 \quad 1.732]^T$ 对 $\hat{x}_A$ 轴旋转 30° , ${}^A P' = ?$

□

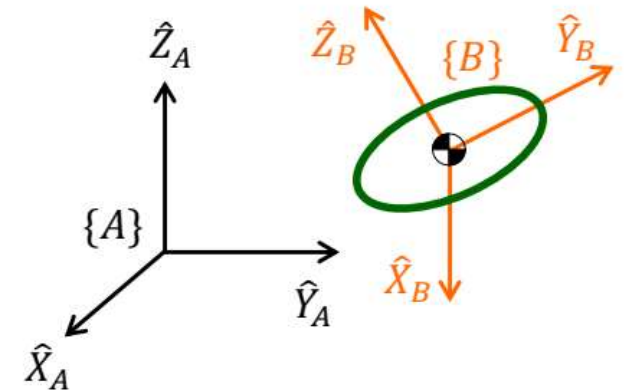


Rotation Matrix -10

□ Rotation matrix 的三种用法

- ◆ 描述一个frame(相对于另一个frame)的姿态

$${}^A_B R = \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix}$$

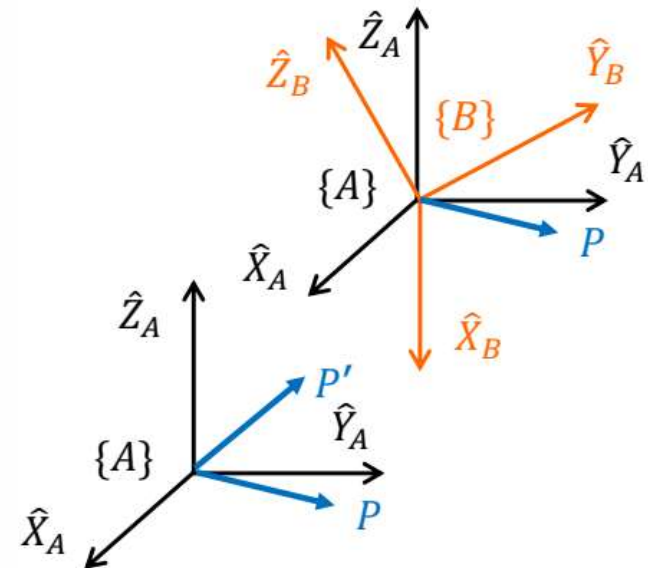


- ◆ 将point由某一个frame的表达换到另一个和此frame仅有相对转动的frame来表达

$${}^A P = {}^A_B R {}^B P$$

- ◆ 将point(vector)在同一个frame中进行转动

$${}^A P' = R(\theta) {}^A P$$



3.1 旋转矩阵与旋转群

◆ 绕x轴旋转的旋转矩阵

$$R(x, \theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

◆ 绕y轴旋转的旋转矩阵

$$R(y, \theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix}$$

◆ 绕z轴旋转的旋转矩阵

$$R(z, \theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

旋转矩阵描述姿态

◆ 思考?

- 这三个旋转矩阵的结构有什么关系, 如果通过记住一个旋转矩阵而得到其它两个?
- 如果绕某一个轴多次旋转, 我们会看到什么结果?
- 如果依次绕x、y、z轴旋转, 旋转矩阵如何计算?